

## Models of Censored Dependent Variable

### Standard (singly censored) Tobit Model

Consider a latent variable model

$$y_t^* = X_t\beta + u_t, \quad u_t \sim i.i.d.(0, \sigma^2)$$

where observations on regressors  $X_t$  are fully available, but observations on the dependent variable are censored<sup>1</sup>:

$$y_t = \begin{cases} y_t^* & \text{if } y_t^* > 0 \\ 0 & \text{if } y_t^* \leq 0 \end{cases}$$

This is called the standard (singly censored) *Tobit* model when  $u_t$  has a normal distribution.

Tobin's (1958) model is a regression equation of the ratio of total durable goods expenditure to disposable income on the age of the head of the household and the ratio of liquid assets to disposable income. Another example of early application is Fair's (1978) model of the number of extra-marital affairs on sex, age, number of years married, number of children, education, occupation, degree of religiousness.

### Properties of the OLSE

Suppose that the error term  $u_t$  satisfies the Gauss-Markov assumptions so that the OLS is BLUE if observations on the dependent variable are not censored. Let  $y_t^+ \equiv y_t^* > 0$  denote the positive observed data. One may apply the OLS to estimate the Tobit model by using

- (a) positively observed data only,  $y_t^+ = X_t\beta + \eta_t$
- (b) all observations (including zeros),  $y_t = X_t\beta + \xi_t$

The OLSE of  $\beta$  is biased and inconsistent in both cases. To show this, consider

$$E(y_t^+) \equiv E(y_t^* | y_t^* > 0) = E(X_t\beta + u_t | u_t > -X_t\beta) = X_t\beta + E(u_t | u_t > -X_t\beta) = X_t\beta + E(u_t^+) \neq X_t\beta$$

$$E(y_t) = P(y_t > 0)E(y_t | y_t > 0) + P(y_t = 0)E(y_t | y_t = 0) = [1 - F(-X_t\beta)]E(y_t^+) \equiv P_t^+ E(y_t^+) \neq X_t\beta$$

where  $F$  is the cumulative distribution function of  $u_t$ ,  $E(u_t^+) = E(u_t | u_t > -X_t\beta)$  is the truncated mean of the error term, and  $P_t^+ = 1 - F(-X_t\beta)$  is the probability of being above the limit. This implies that error terms in the regression equations in (a) and (b) above do not have zero means, and hence OLSE is biased.

Remark: Computation of the truncated mean  $E(u_t^+) = E(u_t | u_t > -X_t\beta)$  uses the truncated density function of  $u_t$ , which is given by

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<sup>1</sup> If observations on the regressors are also censored, then the model is called the truncated data model.

$$f(u_t|u_t > c) = \begin{cases} \frac{f(u_t)}{1 - F(c)} & \text{if } u_t > c \\ 0 & \text{otherwise} \end{cases}$$

where  $c = -X_t\beta$  and  $f$  is the probability density function of  $u_t$ . The mean of this distribution is computed by the usual formula

$$E(u_t|u_t > c) = \int_c^{\infty} z \frac{f(z)}{1 - F(c)} dz$$

### *Marginal Effects of Regressors*

We can consider several effects of a regressor: the marginal effect on the means of the latent dependent variable  $E(y_t^*)$ , of the censored dependent variable  $E(y_t)$ , of the positive dependent variable  $E(y_t^+)$ , of the negative dependent variable  $E(y_t^-)$ , and of the probability of being above the limit  $P_t^+$ .

$$\begin{aligned} \frac{\partial E(y_t^*)}{\partial X_{tk}} &= \beta_k, & \frac{\partial E y_t^+}{\partial X_{tk}} &= \beta_k + \frac{\partial E(u_t^+)}{\partial X_{tk}}, & \frac{\partial P_t^+}{\partial X_{tk}} &= f(-X_t\beta)\beta_k \\ \frac{\partial E(y_t)}{\partial X_{tk}} &= P_t^+ \frac{\partial E y_t^+}{\partial X_{tk}} + E y_t^+ \frac{\partial P_t^+}{\partial X_{tk}} \end{aligned}$$

The last expression shows the decomposition of the marginal effect of a regressor on the expected censored dependent variable  $y_t$  into two components. This decomposition is suggested in Tobin (1958) and worked out in detail by McDonald and Moffitt (1980). In the example of household expenditure on consumer durables, the first term captures the effect of the regressor on the expenditures of those who had positive expenditures. Tobin called the first term “the size of non-limit responses.” The second term captures the effect of the regressor on the change of the households which have a positive expenditure, either moving from positive expenditure to zero expenditure, or vice versa. If a household moves from zero to positive expenditure group, its expected expenditure is  $E(y_t^+)$ .

The marginal effects of a regressor are not constant except for the effect on the mean of the latent variable. They are functions of the values of all regressors. Therefore, the marginal effects can be evaluated at various values of the regressors, and it is a common practice to evaluate them at the sample means of continuous regressors. But, they can also be computed for a set of specific values of the regressors. When a regressor is a discrete variable (like a 0-1 dummy variable), its marginal effect can be evaluated by computing the mean function at value 0 and then at value 1. The difference between the two values is then taken as the marginal effect of a binary regressor. This value of course depends the choice of the values of other regressors.

## Estimation of Censored Model

### (1) Maximum Likelihood Estimator

Observations are either the event of  $y_t^* \leq 0$ , or the value of  $y_t^*$  when  $y_t^* > 0$ . Define a binary dummy variable for censoring

$$d_t = \begin{cases} 0 & \text{if } y_t = 0 \quad (y_t^* \leq 0) \\ 1 & \text{if } y_t > 0 \quad (y_t^* > 0) \end{cases}$$

The single likelihood function is either a discrete probability  $F(-X_t\beta) = P(y_t^* \leq 0)$  for  $y_t = 0$ , or the density  $f(y_t^*)$  for  $y_t > 0$ :

$$L_t(y_t) = F(-X_t\beta)^{1-d_t} f(y_t - X_t\beta)^{d_t}$$

The likelihood function and the log likelihood function for all observations are

$$L = \prod_{t=1}^n \left( F(-X_t\beta) \right)^{1-d_t} \left( f(y_t - X_t\beta) \right)^{d_t}$$

$$\ln L = \sum_{t=1}^n \left( (1-d_t) \ln F(-X_t\beta) + d_t \ln f(y_t - X_t\beta) \right)$$

The MLE of the parameters  $\theta$  is the root of the gradient vector

$$\frac{\partial \ln L}{\partial \theta} = \sum_{t=1}^n \left( (1-d_t) \frac{f(-X_t\beta)}{F(-X_t\beta)} \frac{\partial f(-X_t\beta)}{\partial \theta} + d_t \frac{1}{f(y_t - X_t\beta)} \frac{\partial f(y_t - X_t\beta)}{\partial \theta} \right) = 0$$

and the asymptotic covariance matrix of  $\hat{\theta}$  is  $(n\Psi)^{-1}$ , where  $\Psi$  is the information matrix.

### Tobit Model

Tobin assumed a normal distribution,  $u_t | X_t \sim i.i.n.(0, \sigma^2)$ . Let  $\Phi(z)$  and  $\phi(z)$  be the cdf and pdf of a standard normal random variable. Then,

$$F(-X_t\beta) = P(y_t \leq 0) = P(u_t \leq -X_t\beta) = P\left(\frac{u_t}{\sigma} \leq -\frac{X_t\beta}{\sigma}\right) = \Phi\left(-\frac{X_t\beta}{\sigma}\right) = 1 - \Phi\left(\frac{X_t\beta}{\sigma}\right) \equiv 1 - \Phi_t$$

and the log likelihood function becomes

$$\ell(\beta, \sigma^2) = \sum_{t=1}^n \left\{ (1-d_t) \ln(1 - \Phi_t) + d_t \left[ -(1/2) \ln(\sigma^2) - (1/2) \ln(2\pi) - (y_t - X_t\beta)^2 / 2\sigma^2 \right] \right\}$$

Amemiya(1973) proved that this Tobit log-likelihood function is not globally concave with respect to  $\beta$  and  $\sigma^2$ . Hence, the Newton-Raphson, or the method of scoring, or the alternative iterative algorithm to solve

for  $\beta$  and  $\sigma^2$  require their consistent initial estimators<sup>1</sup>.

Let  $\Phi_t = \Phi(X_t\beta/\sigma)$  and  $\lambda_t = \Phi_t/(1 - \Phi_t)$ . The gradient vectors with respect to  $\beta$  and  $\sigma^2$  are

$$g(\beta) = \frac{\partial \ell(\beta, \sigma^2)}{\partial \beta} = \sum_{t=1}^n \left\{ -(1-d_t) \frac{\lambda_t}{\sigma} X_t' + d_t \frac{1}{\sigma^2} X_t' (y_t - X_t\beta) \right\} = 0$$

$$g(\sigma^2) = \frac{\partial \ell(\beta, \sigma^2)}{\partial \sigma^2} = \sum_{t=1}^n \left\{ (1-d_t) \frac{1}{2\sigma^3} \lambda_t X_t\beta + d_t \left( -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (y_t - X_t\beta)^2 \right) \right\} = 0$$

Equation for  $g(\sigma^2)$  can be further simplified by using the first order condition of  $g(\beta)=0$ . Transpose  $g(\beta)=0$  equation and post-multiply  $\beta/2\sigma^2$  to get

$$\sum_{t=1}^n (1-d_t) \frac{\lambda_t}{2\sigma^3} X_t\beta = \sum_{t=1}^n d_t \frac{1}{2\sigma^4} (y_t - X_t\beta) X_t\beta$$

Substitution of this expression into  $g(\sigma^2)$  equation leads to

$$\sum_{t=1}^n d_t \left( -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (y_t - X_t\beta) y_t \right) = \frac{1}{2\sigma^4} \sum_{t=1}^n d_t \left( -\sigma^2 + (y_t - X_t\beta) y_t \right) = 0$$

Therefore, the MLE of  $\sigma^2$  is given by

$$\hat{\sigma}^2 = \frac{\sum_{t=1}^n d_t (y_t - X_t\hat{\beta}) y_t}{\sum_{t=1}^n d_t}$$

This shows that the estimator of  $\sigma^2$  given the estimator of  $\beta$  uses only the samples with positive dependent variable.

The elements of Hessian matrix are

$$\begin{aligned} \frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \beta \partial \beta'} &= - \sum_{t=1}^n \left\{ (1-d_t) \frac{1}{\sigma^3} \lambda_t (\sigma \lambda_t - X_t\beta) X_t' X_t + d_t \frac{1}{\sigma^2} X_t' X_t \right\} \\ &= - \sum_{t=1}^n \frac{1}{\sigma^2} \left\{ (1-d_t) \lambda_t (\lambda_t - X_t\alpha) X_t' X_t + d_t X_t' X_t \right\} \end{aligned}$$

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<sup>1</sup>Amemiya proposed a simple procedure to find consistent initial estimators by using only the sample of positive values of  $y_t^*$ . Lee extended the method to use both positive and zero observations.

$$\begin{aligned}\frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \beta \partial \sigma^2} &= - \sum_{t=1}^n \left\{ (1-d_t) \frac{1}{2\sigma^5} \lambda_t [(X_t \beta)^2 - \sigma^2 - \sigma \lambda_t X_t \beta] X_t' + d_t \frac{1}{\sigma^4} X_t' (y_t - X_t \beta) \right\} \\ &= - \sum_{t=1}^n \frac{1}{2\sigma^3} \left\{ (1-d_t) \lambda_t [(X_t \alpha)^2 - 1 - \lambda_t X_t \alpha] X_t' + \frac{2}{\sigma} d_t X_t' (y_t - X_t \beta) \right\}\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \sigma^2 \partial \sigma^2} &= \sum_{t=1}^n \left\{ (1-d_t) \frac{1}{4\sigma^7} \lambda_t X_t \beta [(X_t \beta)^2 - 3\sigma^2 - \sigma \lambda_t X_t \beta] + d_t \left( \frac{1}{2\sigma^4} - \frac{1}{\sigma^6} (y_t - X_t \beta)^2 \right) \right\} \\ &= \sum_{t=1}^n \frac{1}{4\sigma^4} \left\{ (1-d_t) \lambda_t X_t \alpha [(X_t \alpha)^2 - 3 - \lambda_t X_t \alpha] + 2d_t - \frac{4}{\sigma^2} d_t (y_t - X_t \beta)^2 \right\}\end{aligned}$$

The expressions of these elements can be simplified by using the first order conditions. From the first order condition  $g(\beta)=0$ , we have

$$\sum_{t=1}^n \frac{1}{\sigma} d_t X_t' (y_t - X_t \beta) = \sum_{t=1}^n (1-d_t) \lambda_t X_t'$$

Substituting this into  $\partial^2 \ell / \partial \beta \partial \sigma^2$ , we can write

$$\begin{aligned}\frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \beta \partial \sigma^2} &= - \sum_{t=1}^n \frac{1}{2\sigma^3} \left\{ (1-d_t) \lambda_t [(X_t \alpha)^2 - 1 - \lambda_t X_t \alpha] X_t' + \frac{2}{\sigma} d_t X_t' (y_t - X_t \beta) \right\} \\ &= - \sum_{t=1}^n \frac{1}{2\sigma^3} \left\{ (1-d_t) \lambda_t [(X_t \alpha)^2 - 1 - \lambda_t X_t \alpha] X_t' + 2(1-d_t) \lambda_t X_t' \right\} \\ &= - \sum_{t=1}^n \frac{1}{2\sigma^3} (1-d_t) \lambda_t [(X_t \alpha)^2 + 1 - \lambda_t X_t \alpha] X_t'\end{aligned}$$

From the first order condition  $g(\sigma^2)=0$  we have

$$\sum_{t=1}^n \frac{1}{\sigma^2} d_t (y_t - X_t \beta)^2 = \sum_{t=1}^n [d_t - (1-d_t) \lambda_t X_t \alpha]$$

Substituting this into  $\partial^2 \ell / \partial \sigma^2 \partial \sigma^2$  we can derive

$$\begin{aligned}\frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \sigma^2 \partial \sigma^2} &= \sum_{t=1}^n \frac{1}{4\sigma^4} \left\{ (1-d_t) \lambda_t X_t \alpha [(X_t \alpha)^2 - 3 - \lambda_t X_t \alpha] + 2d_t - \frac{4}{\sigma^2} d_t (y_t - X_t \beta)^2 \right\} \\ &= \sum_{t=1}^n \frac{1}{4\sigma^4} \left\{ (1-d_t) \lambda_t X_t \alpha [(X_t \alpha)^2 - 3 - \lambda_t X_t \alpha] + 2d_t - 4[d_t - (1-d_t) \lambda_t X_t \alpha] \right\} \\ &= \sum_{t=1}^n \frac{1}{4\sigma^4} \left\{ (1-d_t) \lambda_t X_t \alpha [(X_t \alpha)^2 + 1 - \lambda_t X_t \alpha] - 2d_t \right\}\end{aligned}$$

The asymptotic covariance matrix of the MLE is computed by the inverse of the information matrix  $\Psi$ , where

$$\Psi = \text{plim}_{n \rightarrow \infty} \left( -\frac{1}{n} H(\beta, \sigma^2) \right)$$

When the probability limit is taken, one must take into account the fact that the number of zero and positive observations are also determined randomly. That is, dummy variable  $d_t$  is a random variable.

$$\begin{aligned} \text{plim}_{n \rightarrow \infty} \left( -\frac{1}{n} \frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \beta \partial \beta'} \right) &= \text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n \frac{1}{\sigma^2} \left\{ (1 - \Phi_t) \lambda_t (\lambda_t - X_t \alpha) X_t' X_t + \Phi_t X_t' X_t \right\} \\ &= \text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n \frac{1}{\sigma^2} \left\{ \phi_t (\lambda_t - X_t \alpha) + \Phi_t \right\} X_t' X_t \equiv \text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n a_t X_t' X_t \\ \text{plim}_{n \rightarrow \infty} \left( -\frac{1}{n} \frac{\partial^2 \ell(\beta, \sigma^2)}{\partial \beta \partial \sigma^2} \right) &= \text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n \frac{\phi_t}{2\sigma^3} \left[ (X_t \alpha)^2 + 1 - \lambda_t X_t \alpha \right] X_t' \equiv \text{plim}_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n b_t X_t' \\ \text{plim}_{n \rightarrow \infty} \left( -\frac{1}{n} \frac{\partial^2 \ell}{\partial \sigma^2 \partial \sigma^2} \right) &= \text{plim}_{n \rightarrow \infty} \frac{1}{n} \left( -\sum_{t=1}^n \frac{1}{4\sigma^4} \left\{ \phi_t X_t \alpha \left[ (X_t \alpha)^2 + 1 - \lambda_t X_t \alpha \right] - 2\Phi_t \right\} \right) \end{aligned}$$

Collecting all elements in the information matrix, we can compute the asymptotic covariance of MLE of  $\beta$  and  $\sigma^2$  by

$$V = \text{cov} \begin{pmatrix} \hat{\beta} \\ \hat{\sigma}^2 \end{pmatrix} = \begin{pmatrix} \sum a_t X_t' X_t & \sum b_t X_t' \\ \sum b_t X_t & \sum c_t \end{pmatrix}^{-1}$$

where the summation is for *entire sample*, and

$$\begin{aligned} a_t &= -\tau^2 [(X_t \alpha - \lambda_t) \phi_t - \Phi_t] \\ b_t &= 2\tau^3 [(X_t \alpha - \lambda_t) X_t \alpha \phi_t + \phi_t] \\ c_t &= -4\tau^4 [(X_t \alpha - \lambda_t) (X_t \alpha)^2 \phi_t + X_t \alpha \phi_t - 2\Phi_t] \end{aligned}$$

The log-likelihood function can be written in transformed parameters  $\alpha = \beta / \sigma$  and  $\tau = 1/\sigma$  instead of  $\beta$  and  $\sigma^2$ :

$$\ell(\alpha, \tau) = \sum_t \left\{ (1 - d_t) \ln(1 - \Phi_t) + d_t \ln \tau - (d_t/2) \ln(2\pi) - d_t (\tau y_t - X_t \alpha)^2 / 2 \right\}$$

where  $\Phi_t = \Phi(X_t \beta / \sigma) = \Phi(X_t \alpha)$ . Olsen(1978) showed that this log-likelihood function is globally concave with respect to the transformed parameters  $\alpha$  and  $\tau$ . Therefore, the log-likelihood function in terms of  $\alpha$  and  $\tau$  can be maximized by finding the roots of its gradient vectors, starting from any arbitrary initial values of  $\alpha$  and  $\tau$ . Estimators of  $\beta$  and  $\sigma^2$  can then be recovered from the estimates of  $\alpha$  and  $\tau$ .

The gradient vectors of the log likelihood function in terms of  $\alpha$  and  $\tau$  are

$$g(\alpha) = \frac{\partial \ell(\alpha, \tau)}{\partial \alpha} = \sum_{t=1}^n \left\{ -(1-d_t) \lambda_t X_t' + d_t (\tau y_t - X_t \alpha) X_t' \right\} = 0$$

$$g(\tau) = \frac{\partial \ell(\alpha, \tau)}{\partial \tau} = \sum_{t=1}^n \left\{ (1-d_t)/\tau - d_t (\tau y_t - X_t \alpha) y_t \right\} = 0$$

where  $\lambda_t$  is now expressed in  $\alpha$ :  $\lambda_t = \Phi(X_t \alpha) / (1 - \Phi(X_t \alpha))$ . Note that  $g(\tau) = 0$  can be written as a quadratic function in  $\tau$ ,  $a\tau^2 - b\tau - c = 0$ , and the estimator of  $\tau$  is its positive root

$$\hat{\tau} = \frac{b + \sqrt{b^2 + 4ac}}{2a}$$

where

$$a = \sum d_t y_t^2, \quad b = \sum d_t (X_t \alpha) y_t, \quad c = \sum (1-d_t)$$

The elements of Hessian matrix are

$$\frac{\partial^2 \ell(\alpha, \tau)}{\partial \alpha \partial \alpha'} = - \sum_{t=1}^n \left\{ (1-d_t) \lambda_t (\lambda_t - X_t \alpha) X_t' X_t + d_t X_t' X_t \right\}$$

$$\frac{\partial^2 \ell(\alpha, \tau)}{\partial \alpha \partial \tau} = \sum_{t=1}^n d_t X_t' y_t$$

$$\frac{\partial^2 \ell(\alpha, \tau)}{\partial \tau^2} = \sum_{t=1}^n \left\{ -(1-d_t) \frac{1}{\tau^2} - d_t y_t^2 \right\}$$

The Hessian matrix can be written as

$$H = - \begin{pmatrix} \sum d_t X_t' X_t & - \sum d_t X_t' y_t \\ - \sum d_t y_t X_t & \sum d_t y_t^2 \end{pmatrix} - \begin{pmatrix} \sum (1-d_t) \lambda_t (\lambda_t - X_t \alpha) X_t' X_t & 0 \\ 0 & \sum (1-d_t) / \tau^2 \end{pmatrix}$$

For  $H$  to be positive definite, its all principal minors must have a positive sign. Let  $y_2$  and  $X_2$  be the matrices of  $y_t$  and  $X_t$  for  $y_t^* > 0$ , and  $X_1$  be the matrix of  $X_t$  for  $y_t^* \leq 0$ . Then, the Hessian matrix can be written as

$$H = - \begin{pmatrix} X_2' X_2 & - X_2' y_2 \\ - y_2' X_2 & y_2' y_2 \end{pmatrix} - \begin{pmatrix} X_1' D X_1 & 0 \\ 0 & N_2 / \tau^2 \end{pmatrix}$$

where  $D = \text{diag}(\lambda_t (\lambda_t - X_t \alpha))$  for  $y_t^* \leq 0$  (i.e.,  $d_t = 0$ ). For the first matrix, it is clear that any principal minors of  $X_2' X_2$  takes a positive sign and the determinant of the entire first matrix is

$$\begin{vmatrix} X_2' X_2 & - X_2' y_2 \\ - y_2' X_2 & y_2' y_2 \end{vmatrix} = |X_2' X_2| \cdot |y_2' y_2 - y_2' X_2 (X_2' X_2)^{-1} X_2' y_2| > 0$$

where the second determinant on the right hand side is just the sum of squared residuals of a linear regression of  $y_2$  on  $X_2$ . For the second matrix, we need to show that the diagonal matrix  $D$  is positive definite, that is, to show  $\lambda_t - X_t \alpha > 0$  for all  $t$  such that  $y_t^* \leq 0$ . Let  $Z$  be a standard normal random variable. We have

shown earlier that

$$\lambda_t = \frac{\phi(X_t \alpha)}{1 - \Phi(X_t \alpha)} = E(Z | Z > X_t \alpha)$$

so that

$$\lambda_t - X_t \alpha = E(Z - X_t \alpha | Z > X_t \alpha) > 0$$

The last inequality follows from the fact that  $Z - X_t \alpha$  is a non-negative random variable conditional on  $Z > X_t \alpha$ . Therefore, the Hessian matrix is negative definite, and hence the log-likelihood function is concave in  $\alpha$  and  $\tau$ . The global concavity of the log-likelihood function implies that we can start with any estimators of  $\alpha$  and  $\tau$  in the standard Newton-Rapson or the score method iterative procedure.

## (2) EM Estimator

Define a new dependent variable that replaces the censored value with its conditional expectation

$$\tilde{y}_t = \begin{cases} y_t^* & \text{if } y_t^* > 0 \\ E y_t^- & \text{if } y_t^* \leq 0 \end{cases}$$

where

$$E y_t^- \equiv E(y_t^* | y_t = 0) = E(X_t \beta + u_t | u_t < -X_t \beta) = X_t \beta + E(u_t | u_t < -X_t \beta) = X_t \beta - \sigma [\phi_t / (1 - \Phi_t)]$$

The iterative EM algorithm follows the following procedure

- (i) Choose initial consistent estimators of  $\beta$  and  $\sigma^2$ , and compute  $E y_t^-$ .
- (ii) Compute the new estimator of  $\beta$  by the OLS:  $\hat{\beta} = (X'X)^{-1}X'\tilde{y}$
- (iii) Compute the new estimator of  $\sigma^2$  by using the new  $\beta$  from (ii) and non-censored data

$$\hat{\sigma}^2 = \frac{\sum_{t=1}^n d_t (y_t - X_t \hat{\beta}) y_t}{\sum_{t=1}^n d_t}$$

- (iv) repeat the procedure until convergence.

The EM algorithm can also be written in terms of parameters  $\alpha$  and  $\tau$ . Define

$$\tilde{y}_t = \begin{cases} \tau y_t^* & \text{if } y_t^* > 0 \\ \tau E y_t^- & \text{if } y_t^* \leq 0 \end{cases}$$

where

$$\tau E y_t^- = X_t \alpha - \lambda_t, \quad \lambda_t = \phi_t / (1 - \Phi_t)$$

The maximization step of the EM algorithm gives

$$\hat{\alpha} = (X'X)^{-1}X'\tilde{y}$$

and the estimator of  $\tau$  as the solution to

$$\tau^2 \sum_{N_2} (y_t^*)^2 - \tau \sum_{N_2} (X_t \alpha) y_t^* - N_2 = 0 \quad \Rightarrow \quad a\tau^2 - b\tau - c = 0$$

which is a quadratic function in  $\tau$ . Since  $\tau$  is positive, we take the positive root

$$\hat{\tau} = \frac{b + \sqrt{b^2 + 4ac}}{2a}$$

The iterative procedure starts with any initial values of  $\alpha$  and  $\tau > 0$ . Compute  $\lambda_i$  and  $X_i \alpha - \lambda_i$  to construct the vector  $\tilde{y}$ , and estimate  $\alpha$ . The estimate of  $\tau$  is then computed as the positive root of the quadratic equation specified above. This process is repeated until convergence is achieved.

[3] Heckman's two-step method

This method is based on a regression equation for only the non-censored sample

$$y_t^* = X_t \beta + (\phi_t / \Phi_t) \sigma + \eta_t, \quad y_t^* = y_t > 0$$

or for the entire sample

$$y_t = (\Phi_t X_t) \beta + \phi_t \sigma + \eta_t$$

Parameters  $\beta$  and  $\sigma$  can be estimated from these equations by OLS if we have observations on  $\phi_t$  and  $\Phi_t$ . Heckman suggested to estimate them from a probit model of the entire sample, by setting the observed dependent  $y_t = 1$  for  $y_t^* > 0$ .

### Truncated Model

This is the case where observations on both dependent and independent variables are not available for a sub-population. The Tobit model becomes a truncated model if the sample of negative  $y_t^*$  is not used in the estimation of the population parameter  $\beta$ . The least squares estimator of  $\beta$  is biased and inconsistent in the case of truncated data set, i.e., using the observations of positive  $y_t^*$  only. Consider a regression equation

$$y_t^* = X_t\beta + u_t^+, \quad y_t^* > 0$$

When  $u_t$  is a normal random variable, the moments of  $u_t^+$  are

$$E(u_t^+ | y_t^* > 0) = \frac{\phi_t}{\Phi_t} \sigma$$

$$\text{var}(u_t^+ | y_t^* > 0) = \sigma^2 [1 - \kappa_t(\kappa_t + X_t\alpha)], \quad \kappa_t = \phi_t / \Phi_t$$

The LSE of  $\beta$  is biased and inconsistent because of the nonzero mean of the error term. The regression error term is also heteroscedastic.

A consistent estimator of the population parameters can be obtained by the MLE. The truncated density function of  $y_t^*$  is

$$f(y_t^* | y_t^* > 0) = \begin{cases} \frac{(1/\sigma)\phi((y_t^* - X_t\beta)/\sigma)}{\Phi(X_t\beta/\sigma)} & \text{if } y_t^* > 0 \\ 0 & \text{otherwise} \end{cases}$$

and hence, the log-likelihood function is

$$\ell = -\sum_t \ln(\Phi_t) - \frac{N_2}{2} \ln(2\pi) - \frac{N_2}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_t (y_t^* - X_t\beta)^2$$

where the summation is only over  $N_2$  sample of positive  $y_t^*$ . The gradient vector is

$$\begin{aligned} \frac{\partial \ell}{\partial \beta} &= -\sum_t \frac{\phi_t}{\sigma \Phi_t} X_t' + \frac{1}{\sigma^2} \sum_t (y_t^* - X_t\beta) X_t' \\ \frac{\partial \ell}{\partial \sigma^2} &= \sum_t \frac{X_t\beta \phi_t}{2\sigma^3 \Phi_t} - \frac{N_2}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_t (y_t^* - X_t\beta)^2 \end{aligned}$$

### Incidental Truncation - Type II Tobit Model

In censored and truncated models, observations of the dependent variable is based on the magnitude of the dependent variable itself. What if the observation of the dependent variable depends on the magnitude (sign) of another variable? One of the earliest studies is Gronau's (1973) model:

$$\begin{aligned} w_i &= X_{2i}\gamma + v_i \\ w_i^r &= Z_i\gamma^r + v_i^r \end{aligned}$$

where  $w_i$  is the wage rate offered to individual  $i$  in the market regardless of the number of working hours and  $w_i^r$  is the reservation wage rate of individual  $i$  which is not an observable variable. An individual participates in the labor market and we can observe her market wage if and only if  $w_i^r - w_i > 0$ .

Heckman's (1974) paper is motivated by his observation of the research on married woman's labor supply, including wage rates, hours of work and decision to work. Typical research estimated the labor supply by estimating separately an equation for hours of work and an equation for the probability of work, or estimate the same labor supply function for the working and non-working women assigning zero hour for the non-working women. His model is a structural model which includes the functions of "the probability that a woman works, her hours of work, her observed wage rate, and her asking wage or shadow price of time." The shadow price is now called the reservation wage. An equilibrium condition is that when a woman works her hours of work adjust to equate the market wage and reservation wage.

Heckman specifies the reservation wage  $w^*$  and market wage  $w$  functions as

$$\begin{aligned} w^* &= g(h, w_m, P, A, Z) \\ w &= B(E, S) \end{aligned}$$

where  $h$ =hours of work,  $w_m$ =wage of husband,  $P$ =price of other goods,  $A$ =household's asset income,  $Z$ =the other characteristics such as number of children, education level etc.,  $E$ =job experiences, and  $S$ =Schooling. The endogenous variables are wage rates and hours of work  $h$ . Specify these equations in linear form

$$\begin{aligned} \ln(w^*) &= \beta_0 + \beta_1 h + X_2 \beta_2 + \epsilon \\ \ln(w) &= b_0 + b_1 S + b_2 E + u \end{aligned}$$

where  $(\epsilon, u)$  are i.i.d.  $N(0, \Sigma)$ . Let  $h^*$  be the desired hours of work that satisfies  $\ln(w) = \ln(w^*)$ , i.e.,

$$h^* = \frac{1}{\beta_1} (b_0 + b_1 S + b_2 E - \beta_0 - X_2 \beta_2 + u - \epsilon)$$

A woman works only if  $\ln(w^*(h=0)) < \ln(w)$ , that is, only if  $h^* > 0$ . The reduced form model of Heckman consists of the market wage equation and the desired hours of work equation with observations on wage and hours of work only if  $h^* > 0$ .

$$\ln(w) = \begin{cases} b_0 + b_1 S + b_2 E + u & \text{if } h^* > 0 \\ - & \text{if } h^* \leq 0 \end{cases}$$

We can put his reduced form model in a formal form as

$$\begin{aligned} y_1^* &= X_1\beta_1 + u_1 \\ y_2^* &= X_2\beta_2 + u_2 \end{aligned}$$

where the first equation is the market wage equation, and the second equation is the desired hours of work (sample selection) equation. The error terms  $(u_1, u_2)$  are i.i.d. random vector with zero means and a covariance matrix  $\Sigma = (\sigma_{ij})$ .

The sample selection is determined by

$$y_1 = \begin{cases} y_1^* & \text{if } y_2^* > 0 \\ - & \text{if } y_2^* \leq 0 \end{cases}$$

The observed value  $y_1$  can be positive or negative in general and there are *no observations* on  $y_1^*$  when  $y_2^* \leq 0$ . Some authors specify  $y_1 = 0$  when  $y_2^* \leq 0$ , but it does not really mean that the observed value is zero: it simply states the event of no observation on  $y_1^*$ . That is, observations on the first equation are *truncated*. There is no need for  $X_1$  to be observed if  $y_2^* \leq 0$ , though we assume full observations on  $X_2$ .

Sometimes, it is assumed that only the signs of  $y_2^*$  are observed, i.e., a binary response model for  $y_2^*$ :

$$y_2 = \begin{cases} 0 & \text{if } y_2^* \leq 0 \\ 1 & \text{if } y_2^* > 0 \end{cases}$$

and the sample selection is specified as

$$y_1 = \begin{cases} y_1^* & \text{if } y_2 = 1 \\ - & \text{if } y_2 = 0 \end{cases}$$

This specification makes it clear that the second equation is a binary choice (probit or logit) model. Because of this nature, the variance of  $u_2$  is not identifiable and it is set  $\sigma_{22} = 1$ . The data consists of  $(y_2, y_1)$ .

Some authors treat the first equation as a censored equation with  $y_1 = 0$  when  $y_2^* \leq 0$  and full observations on  $X_1$  are available. An example is an individual's decision on charitable contribution: decide whether to make a contribution or not ( $y_2 = 0$  or 1), and then decides on the amount  $y_1 \geq 0$ . However, the information on  $y_1 = 0$  is completely determined by the second equation and hence it does not add anything to the analysis of the contribution amount.

## Estimation

### (1) Least Squares Estimation

The estimator of  $\beta_1$  from  $y_1 = X_1\beta_1 + \epsilon_1$  using the observed values of  $y_1^*$  is biased as the data is truncated. This bias is called the *sample selection bias*<sup>1</sup>. The source of this bias is same as the case of

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<sup>1</sup> Some authors use the term *sample selection bias* for the case of censored and truncated models. However, the term is used mostly for the case of incidental truncation.

censored and truncated models, i.e.,  $E(y_1) = E(y_1^* | y_2^* > 0) \neq X_1 \beta_1$ . When error terms are distributed as normal, it is straightforward to show

$$\begin{aligned} E(y_1) &= E(y_1^* | y_2^* > 0) = X_1 \beta_1 + (\sigma_{12}/\sigma_2) \kappa_2 \\ \text{var}(y_1) &= \text{var}(y_1^* | y_2^* > 0) = \sigma_{11} - (\sigma_{12}^2/\sigma_{22}) \kappa_2 (\kappa_2 + X_2 \alpha_2) \end{aligned}$$

where

$$\kappa_2 = \frac{\phi(X_2 \beta_2 / \sigma_2)}{\Phi(X_2 \beta_2 / \sigma_2)} = \frac{\phi(X_2 \alpha_2)}{\Phi(X_2 \alpha_2)}, \quad \sigma_2 = \sqrt{\sigma_{22}}$$

These results suggest a heteroscedastic regression model for the observed  $y_1$

$$y_1 = X_1 \beta_1 + (\sigma_{12}/\sigma_2) \kappa_2 + \xi_1, \quad E(\xi_1) = 0, \quad E(\xi_1^2) = \sigma_{11} - (\sigma_{12}^2/\sigma_{22}) \kappa_2 (\kappa_2 + X_2 \alpha_2)$$

The second term in this expression is the source of the bias and inconsistency of the least squares estimator of  $\beta_1$ . Note that  $\sigma_2$  appears always as a combination with  $\beta_2$  in  $\kappa_2$  or as a combination with  $\sigma_{12}$ . Therefore,  $\sigma_2$  is not identifiable and is usually set  $\sigma_2 = 1$ . Note also that, if two error terms are not correlated ( $\sigma_{12} = 0$ ), then the OLSE is a consistent estimator.

Remarks.

- (a) If error terms are uncorrelated ( $\sigma_{12} = 0$ ), then the OLSE of  $\beta_1$  from the regression equation  $y_1 = X_1 \beta_1 + \epsilon_1$  is BLUE if other standard assumptions are valid. That is, there is no sample selection bias if  $\sigma_{12} = 0$ .
- (b)  $E(y_1)$  is a nonlinear function of  $X_2$  as it enters through  $\kappa_2$ . Hence, if one estimates an expanded regression including  $X_2$ ,  $y_1 = X_1 \beta_1 + X_2 \delta_2 + e_1$ , then, the coefficient  $\delta_2$  may turn out to be significant even if  $X_2$  does not belong to the first latent variable equation. Its significance is solely due to the sample selection problem.
- (c) If a common variable is included in  $X_1$  and  $X_2$ , one may make a false conclusion that the proper equation is nonlinear in that variable.

## (2) Heckman's two step estimator

From the mean function  $E(y_1)$ , we write a regression equation

$$y_1 = X_1 \beta_1 + \delta \frac{\phi(X_2 \alpha_2)}{\Phi(X_2 \alpha_2)} + \xi_1$$

where  $\delta = (\sigma_{12}/\sigma_2)$  and  $\xi$  is heteroskedastic. If  $\alpha_2$  is known, we can compute  $\phi(X_2 \alpha_2)/\Phi(X_2 \alpha_2)$  and then estimate  $\beta_1$  and  $\delta$  by the OLS or the GLS (for heteroskedasticity). Heckman's two-step procedure is

first step: estimate  $\alpha_2$  with probit MLE (setting  $\sigma_{22} = 1$  since it is not identifiable)

second step: estimate  $\beta_1$  by the OLS or the GLS

### (3) Maximum Likelihood Estimator

The likelihood function is a joint density of all observations, which is a multiplication of the densities of each observation for a random sample. In this model, we have two sets of observations,  $\{y_2=0, \text{no } y_1\}$  and  $\{y_2=1, y_1=y_1^*\}$ . The probability of observing the first set is just the probability of the event  $\{y_2=0\}$ . The probability of observing the second set is the joint probability of observing  $y_2=1$  and  $y_1=y_1^*$ . This joint probability can be written as  $f(y_1^*|y_2=1)P(y_2=1)$ . Therefore, the likelihood function can be written as

$$L = \prod_{y_{i2}=0} P(y_{i2}=0) \times \prod_{y_{i2}=1} [f(y_{i1}^*|y_{i2}=1)P(y_{i2}=1)] = \prod_{i=1}^n [P(y_{i2}=0)]^{1-y_{i2}} [f(y_{i1}^*|y_{i2}=1)P(y_{i2}=1)]^{y_{i2}}$$

Under the assumptions of normality, the first term can be written as

$$P(y_{i2}=0) = P(y_{i2}^* \leq 0) = P(u_{i2} \leq -X_{i2}\beta_2) = 1 - \Phi(X_{i2}\beta_2/\sigma_2) = 1 - \Phi(X_{i2}\alpha_2)$$

For the second term, it is easier to rewrite it as<sup>1</sup>

$$f(y_{i1}^*|y_{i2}=1)P(y_{i2}=1) = P(y_{i2}=1|y_{i1}^*)f(y_{i1}^*) = P(y_{i2}^* > 0|y_{i1}^*)f(y_{i1}^*)$$

where  $f(y_{i1}^*)$  is the unconditional marginal density of  $y_{i1}^*$ . The conditional probability in the last expression can be found from the conditional distribution of  $y_{i2}^*$  conditional on  $y_{i1}^*$ . Under the normality assumption, this conditional distribution is a normal distribution

$$y_{i2}^*|y_{i1}^* \sim N(\mu_{y_2|y_1}, \sigma_{y_2|y_1}^2)$$

$$\mu_{y_2|y_1} = X_{i2}\beta_2 + (\sigma_{12}/\sigma_{11})(y_{i1}^* - X_{i1}\beta_1)$$

$$\sigma_{y_2|y_1}^2 = \sigma_{22} - \sigma_{21}\sigma_{11}^{-1}\sigma_{12}$$

Therefore,

$$f(y_{i1}^*) = \frac{1}{\sigma_1} \phi\left(\frac{y_{i1}^* - X_{i1}\beta_1}{\sigma_1}\right)$$

and

$$P(y_{i2}^* > 0|y_{i1}^*) = 1 - P(y_{i2}^* \leq 0|y_{i1}^*) = \Phi\left(\frac{\mu_{y_2|y_1}}{\sigma_{y_2|y_1}}\right) = \Phi\left(\frac{X_{i2}\alpha_2 + (\rho/\sigma_1)(y_{i1}^* - X_{i1}\beta_1)}{(1-\rho^2)^{1/2}}\right)$$

where  $\sigma_1 = \sqrt{\sigma_{11}}$  and  $\rho = \sigma_{12}/(\sigma_{11}\sigma_{22})^{1/2}$ . Putting all elements together, we can write the likelihood

$$L = \prod_{i=1}^n \left[1 - \Phi(X_{i2}\alpha_2)\right]^{1-y_{i2}} \times \prod_{i=1}^n \left( \Phi\left(\frac{\mu_{y_2|y_1}}{\sigma_{y_2|y_1}}\right) \times \frac{1}{\sigma_1} \phi\left(\frac{y_{i1}^* - X_{i1}\beta_1}{\sigma_1}\right) \right)^{y_{i2}}$$

and

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<sup>1</sup>This uses the fact that  $f(z_1, z_2) = f(z_1|z_2)f(z_2) = f(z_2|z_1)f(z_1)$

$$\begin{aligned} \ln L = & \sum_{i=1}^n (1-y_{i2}) \ln [1 - \Phi(X_{i2}\alpha_2)] + \sum_{i=1}^n y_{i2} \ln \left\{ \frac{1}{\sigma_1} \phi \left( \frac{y_{i1} - X_{i1}\beta_1}{\sigma_1} \right) \right\} \\ & + \sum_{i=1}^n y_{i2} \ln \Phi \left( \frac{X_{i2}\alpha_2 + (\rho/\sigma_1)(y_{i1} - X_{i1}\beta_1)}{(1-\rho^2)^{1/2}} \right) \end{aligned}$$

where we used the observed value  $y_{i1}$  in the second and third terms because these terms represent the samples with observed  $y_1^*$ . Note that the first two terms are similar to the terms that appear in the probit model, but the last term is a new term for the model under consideration. Since  $\sigma_2$  appears in the likelihood function only in  $\alpha_2$  and  $\rho$ ,  $\sigma_2$  is not identifiable, and hence can be set to  $\sigma_2 = 1$ .

## Appendix

## Type II Tobit Model - Sample Selection bias

Model:

$$y_1^* = X_1\beta_1 + u_1$$

$$y_2^* = X_2\beta_2 + u_2$$

Sample selection:

$$y_1 = y_1^* \text{ if } y_2^* > 0$$

and no observations on  $y_1^*$  and  $X_1$  if  $y_2^* \leq 0$ .

*Theorem.* Let  $z_1$  and  $z_2$  be jointly normal with means  $\mu_1$  and  $\mu_2$ , and covariance matrix  $\Sigma = (\sigma_{ij})$ . Then, the conditional distribution of  $z_1$  given  $z_2$  is normal

$$z_1|z_2 \sim N[\mu_1 + (\sigma_{12}/\sigma_{22})(z_2 - \mu_2), \quad \sigma_{11} - \sigma_{12}\sigma_{22}^{-1}\sigma_{21}]$$

This can be expressed as

$$z_1 = \mu_1 + (\sigma_{12}/\sigma_{22})(z_2 - \mu_2) + \eta_1$$

where

$$E(\eta_1) = 0, \quad \text{var}(\eta_1) = \sigma_{11} - \sigma_{12}\sigma_{22}^{-1}\sigma_{21}, \quad \text{cov}(z_2, \eta_1) = 0$$

*Theorem.* Suppose that  $u_1$  and  $u_2$  are jointly normal with zero means and covariance matrix  $\Sigma = (\sigma_{ij})$ . Then,

$$E(y_1) = E(y_1^* | y_2^* > 0) = X_1\beta_1 + (\sigma_{12}/\sigma_2)\kappa_2$$

$$\text{var}(y_1) = \text{var}(y_1^* | y_2^* > 0) = \sigma_{11} - \sigma_{12}^2\sigma_{22}^{-1}(\kappa_2^2 + \kappa_2 X_2\alpha_2)$$

where  $\sigma_2 = \sqrt{\sigma_{22}}$  and

$$\kappa_2 = \frac{\phi(X_2\beta_2/\sigma_2)}{\Phi(X_2\beta_2/\sigma_2)} = \frac{\phi(X_2\alpha_2)}{\Phi(X_2\alpha_2)}$$

*Proof.* Joint normality of  $u_1$  and  $u_2$  implies joint normality of  $y_1^*$  and  $y_2^*$ , which have means  $X_1\beta_1$  and  $X_2\beta_2$ , respectively, and covariance matrix  $\Sigma$ . Therefore, by the theorem above, we can write

$$y_1^* = X_1\beta_1 + (\sigma_{12}/\sigma_{22})(y_2^* - X_2\beta_2) + \eta_1$$

where

$$\text{var}(\eta_1) = \sigma_{11} - \sigma_{12}\sigma_{22}^{-1}\sigma_{21}, \quad \text{cov}(y_2^*, \eta_1) = 0$$

Therefore,

$$E(y_1) = E(y_1^* | y_2^* > 0) = X_1\beta_1 + (\sigma_{12}/\sigma_{22})[E(y_2^* | y_2^* > 0) - X_2\beta_2] = X_1\beta_1 + (\sigma_{12}/\sigma_2)\kappa_2$$

where the last equality is due the result

$$E(y_2^* | y_2^* > 0) = X_2\beta_2 + \sigma_2\kappa_2$$

The conditional variance is found from

$$\text{var}(y_1^* | y_2^* > 0) = (\sigma_{12}/\sigma_{22})^2 \text{var}(y_2^* | y_2^* > 0) + \text{var}(\eta_1)$$

and

$$\text{var}(y_2^* | y_2^*) = \sigma_{22}[1 - \kappa_2(\kappa_2 + X_2\alpha)]$$

## Appendix

### Moments of Truncated Distribution

Let  $f(x)$  and  $F(x)$  be the pdf and cdf of  $X$ . It was shown earlier that a distribution truncated from below by a constant  $c$  is a conditional distribution  $f(x|X>c)$ :

$$f(x|X>c) = \begin{cases} \frac{f(x)}{1-F(c)} & \text{if } x > c \\ 0 & \text{otherwise} \end{cases}$$

The mean and variance of this distribution are computed by the usual formulae

$$E(X|X>c) = \int_c^{\infty} x \frac{f(x)}{1-F(c)} dx, \quad \text{Var}(X|X>c) = E(X^2|X>c) - [E(X|X>c)]^2$$

Consider for an example a normal random variable  $X \sim N(\mu, \sigma^2)$ . We know that  $Z = (X-\mu)/\sigma$  is distributed as a standard normal  $N(0, 1)$ , and

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} = \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2} = \frac{1}{\sigma} \phi(z), \quad z = \frac{x-\mu}{\sigma}$$

$$F(c) = P(X \leq c) = P\left(\frac{X-\mu}{\sigma} \leq \frac{c-\mu}{\sigma}\right) = P(Z \leq b) = \Phi(b), \quad b = \frac{c-\mu}{\sigma}$$

Therefore,

$$\begin{aligned} E(X|X>c) &= \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{1-F(c)} \int_c^{\infty} x e^{-(x-\mu)^2/2\sigma^2} dx = \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{1-\Phi(b)} \int_b^{\infty} (\mu + \sigma z) e^{-z^2/2} (\sigma dz) \\ &= \frac{\mu}{1-\Phi(b)} [1-\Phi(b)] + \frac{\sigma}{1-\Phi(b)} \frac{1}{\sqrt{2\pi}} e^{-b^2/2} = \mu + \sigma \frac{\phi(b)}{1-\Phi(b)} \equiv \mu + \sigma \lambda^+(b) \end{aligned}$$

where  $\lambda^+(b) \equiv \phi(b)/[1-\Phi(b)]$  is the hazard function, which is also called the *inverse Mills ratio*. The conditional variance can be derived by first computing the conditional second moment

$$\begin{aligned} E(X^2|X>c) &= \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{1-F(c)} \int_c^{\infty} x^2 e^{-(x-\mu)^2/2\sigma^2} dx = \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{1-\Phi(b)} \int_b^{\infty} (\mu + \sigma z)^2 e^{-z^2/2} (\sigma dz) \\ &= \frac{\mu^2}{1-\Phi(b)} \frac{1}{\sqrt{2\pi}} \int_b^{\infty} e^{-z^2/2} dz + \frac{2\sigma\mu}{1-\Phi(b)} \frac{1}{\sqrt{2\pi}} \int_b^{\infty} z e^{-z^2/2} dz + \frac{\sigma^2}{1-\Phi(b)} \frac{1}{\sqrt{2\pi}} \int_b^{\infty} z^2 e^{-z^2/2} dz \\ &= \mu^2 + 2\sigma\mu\lambda(b) + b\sigma^2\lambda(b) + \sigma^2 \end{aligned}$$

where the last integral is obtained by the *integration by parts*, setting  $u=z$ ,  $dv=ze^{-z^2/2}dz$ , and  $v=-e^{-z^2/2}$ :

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_b^{\infty} z^2 e^{-z^2/2} dz &= \frac{1}{\sqrt{2\pi}} \int_b^{\infty} u dv = \frac{1}{\sqrt{2\pi}} uv \Big|_b^{\infty} - \frac{1}{\sqrt{2\pi}} \int_b^{\infty} v du \\ &= b\phi(b) + [1-\Phi(b)] \end{aligned}$$

Now the conditional variance is derived from

$$\begin{aligned} \text{var}(X|X>c) &= E(X^2|X>c) - [E(X|X>c)]^2 = (\mu^2 + 2\sigma\mu\lambda^+(b) + b\sigma^2\lambda^+(b) + \sigma^2) - [\mu + \sigma\lambda^+(b)]^2 \\ &= \sigma^2\{1 - \lambda^+(b)[\lambda^+(b) - b]\} \end{aligned}$$

Note that the conditional mean is greater than the unconditional mean, and the conditional variance is smaller than unconditional variance. Note also that  $\{1 - \lambda^+(b)[\lambda^+(b) - b]\}$  must be positive for any finite  $b$ .

Exercise: Show that, for a normal random variable  $X \sim N(\mu, \sigma^2)$ , and  $b = (c - \mu)/\sigma$ ,

$$\begin{aligned} E(X|X<c) &= \mu + \sigma\lambda^-(b), & \lambda^- &= -\frac{\phi(b)}{\Phi(b)} \\ \text{var}(X|X<c) &= \sigma^2[1 - \lambda^-(b)(\lambda^-(b) - b)] \end{aligned}$$

Exercise: Consider a logistic random variable  $X$  whose cdf is  $F(x) = 1/(1+e^{-x})$  and pdf  $f(x) = F(x)F(-x) = F(x)[1 - F(x)]$ . Show

$$E(X|X<c) = \frac{cF(c) + \ln(1-F(c))}{F(c)}, \quad E(X|X>c) = -\frac{cF(c) + \ln(1-F(c))}{1-F(c)}$$

Exercise: Suppose  $X \sim U(a, b)$ . Find  $E(X|X>c)$  and  $\text{var}(X|X>c)$ . □

#### Expected Value of a Truncated Normal Random Variable

To find the means of the truncated random variables from the truncated density function of  $u_t > -X_t\beta$  and  $u_t < -X_t\beta$ , let  $u_t \sim N(0, \sigma^2)$ . Then,

$$\begin{aligned} f^+(u_t) &= \begin{cases} \frac{(1/\sigma)\phi(u_t/\sigma)}{\Phi(X_t\beta/\sigma)} & \text{if } u_t > -X_t\beta \\ 0 & \text{otherwise} \end{cases} \\ f^-(u_t) &= \begin{cases} \frac{(1/\sigma)\phi(u_t/\sigma)}{1 - \Phi(X_t\beta/\sigma)} & \text{if } u_t \leq -X_t\beta \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Therefore,

$$E(u_t | u_t > -X_t\beta) = \frac{1}{\Phi(X_t\beta/\sigma)} (2\pi\sigma^2)^{-1/2} \int_{-X_t\beta}^{\infty} z e^{-z^2/(2\sigma^2)} dz = \frac{\sigma^2 f(X_t\beta)}{\Phi(X_t\beta/\sigma)} = \frac{\sigma \phi_t}{\Phi_t}$$

$$E(u_t | u_t \leq -X_t\beta) = -\frac{\sigma \phi_t}{1 - \Phi_t}$$

*Expected Value of a Truncated Logistic Random Variable*

Let  $Z$  be a logistic random variable with cdf  $F(z) = 1/(1+e^{-z})$ . The means of  $Z$  truncated at a constant  $c$  are

$$E(Z | Z \leq c) = \frac{1}{F(c)} \int_{-\infty}^c z dF(z), \quad E(Z | Z > c) = \frac{1}{1 - F(c)} \int_c^{\infty} z dF(z)$$

Consider first the following indefinite integration by parts:

$$\begin{aligned} \int z dF(z) &= zF(z) - \int F(z) dz = zF(z) - \left( z + \ln(1 + e^{-z}) \right) = z/(1 + e^{-z}) - (\ln(e^z) + \ln(1 + e^{-z})) \\ &= z/(1 + e^{-z}) - \ln(1 + e^z) = zF(z) + \ln(1 - F(z)) \end{aligned}$$

It can be shown that

$$\lim_{z \rightarrow -\infty} (zF(z) - \ln(1 - F(z))) = \lim_{z \rightarrow \infty} (zF(z) - \ln(1 - F(z))) = 0$$

Hence,

$$E(Z | Z \leq c) = \frac{cF(c) + \ln(1 - F(c))}{F(c)}, \quad E(Z | Z > c) = -\frac{cF(c) + \ln(1 - F(c))}{1 - F(c)}$$